Stochastic optimization
Chance-constrained programming

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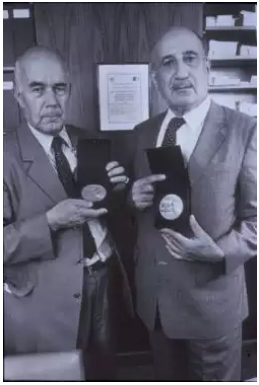
A long story

- Introduced in 1959 by Charnes and Cooper
  `https://dl.acm.org/doi/10.1287/mnsc.6.1.73`
- And also a bit improbable.
- Cooper dropped out of high school to support his family, and became a professional boxer.
- He became an accountant for Eric Louis Kohler, whom he met while hitchhiking.
- Kohler financed his bachelor's degree at the University of Chicago.
- At 26, he enrolled at Columbia University and finished his coursework and dissertation, but never received his PhD due to his dissertation's claim that decision making was not a centralized process.

A long story (cont'd)

- The collaboration with Charnes was however successful, with more than 200 publications, and led to a successful academic career.
- Source: `https://www.informs.org/Explore/History-of-O.R.-Excellence/Biographical-Profiles/Cooper-William-W`

Cooper and Charnes



John von Neumann Theory Prize, 1982 (with Richard J. Duffin)

Mathematical Formulation

$$\begin{aligned} \min_{x \in \mathcal{X}} \quad & \mathbb{E}[f(x, \xi)] \\ \text{s.t.} \quad & P[g_i(x, \xi) \leq 0] \geq 1 - \alpha_i, \quad i = 1, \ldots, m, \end{aligned}$$

where:

- $\xi \in \mathbb{R}^p$ is a random vector with known probability distribution P;
- $x \in \mathbb{R}^n$ is the vector of decision variables;
- $\mathcal{X} \subseteq \mathbb{R}^n$ is a convex deterministic set;
- $f(x, \xi)$ is the cost function;
- $g_i(x, \xi)$ are random constraint functions;
- $\alpha_i \in (0, 1)$ specifies the acceptable risk level for constraint i.

Probabilistic constraints

The probabilistic constraint

$$P[g_i(x, \xi) \leq 0] \geq 1 - \alpha_i$$

ensures that constraint i is satisfied with probability at least $1 - \alpha_i$. Equivalently, the probability of violation

$$P[g_i(x, \xi) > 0] \leq \alpha_i$$

is limited to a user-specified tolerance. Smaller values of α_i produce more conservative, risk-averse solutions.

Individual and joint chance constraints

Two common variants are:

1. **Individual (or separate) chance constraints:** each constraint $g_i(x, \xi) \leq 0$ is required to hold with probability at least $1 - \alpha_i$;
2. **Joint chance constraints:**

$$P[g_i(x, \xi) \leq 0,\ i = 1, \ldots, m] \geq 1 - \alpha,$$

which ensures simultaneous satisfaction of all constraints with a single confidence level $1 - \alpha$.

Toy example

Source: J. Linderoth
(https://jlinderoth.github.io/classes/ie495/lecture22.pdf)

Consider the toy problem

$$\begin{aligned} \min_x \ & x_1 + x_2 \\ \text{s.t. } & \xi_1 x_1 + x_2 \geq 7 \\ & \xi_2 x_1 + x_2 \geq 4 \\ & x_1, x_2 \geq 0. \end{aligned}$$

Instead of requiring that a constraint holds for all the scenarios, we can require that the constraint is satisfied with a given (large) probability.

Chance constraints

1. Separate chance constraints

$$P[\xi_1 x_1 + x_2 \geq 7] \geq 1 - \alpha_1$$
$$P[\xi_2 x_1 + x_2 \geq 4] \geq 1 - \alpha_2$$

2. Joint chance constraint

$$P[\xi_1 x_1 + x_2 \geq 7,\ \xi_2 x_1 + x_2 \geq 4] \geq 1 - \alpha$$

Example: joint chance constraints

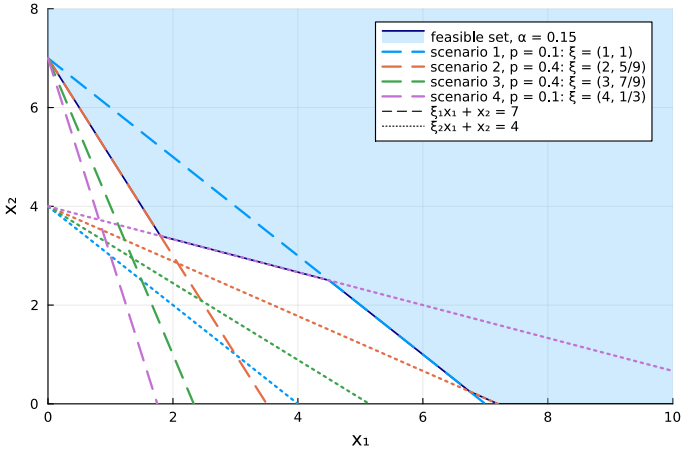
$$P[(\xi_1, \xi_2) = (1, 1)] = 0.1 \quad (1)$$
$$P[(\xi_1, \xi_2) = (2, 5/9)] = 0.4 \quad (2)$$
$$P[(\xi_1, \xi_2) = (3, 7/9)] = 0.4 \quad (3)$$
$$P[(\xi_1, \xi_2) = (4, 1/3)] = 0.1 \quad (4)$$

Assume that $\alpha \in [0.1, 0.2)$, and we have the joint constraint

$$P[\xi_1 x_1 + x_2 \geq 7,\ \xi_2 x_1 + x_2 \geq 4] \geq 1 - \alpha$$

We then have to satisfy the constraints in scenarios (2) and (3), and in either (1) or (4).

Example: frontiers of constraints



Feasible set

Feasible set:

$$K_1(\alpha) = \{x \mid P[g_i(x, \xi) \leq 0] \geq 1 - \alpha_i,\ i = 1, \ldots, m\}$$

$K_1(\alpha)$ is not necessarily convex, even if $g_i(x, \xi)$, $i = 1, \ldots, m$, are linear in x: the joint constraint of the example yields a union of two polyhedra.

Quasi-concavity

Definition (Quasi-Concavity)

A function $f : S \to \mathbb{R}$, where $S \subseteq \mathbb{R}^n$ is a convex set, is **quasi-concave** if for all $\boldsymbol{x}, \boldsymbol{y} \in S$ and all $\lambda \in [0, 1]$:

$$f(\lambda \boldsymbol{x} + (1 - \lambda)\boldsymbol{y}) \geq \min\{f(\boldsymbol{x}), f(\boldsymbol{y})\}$$

Definition

The probability measure P defined on the Borel sets of $\mathbb{R}^n$ is said to be quasi-concave, if for any convex subsets A, B of $\mathbb{R}^n$, and any $\lambda \in [0, 1]$,

$$P[(1 - \lambda)A + \lambda B] \geq \min\{P[A], P[B]\}.$$

(See Prékopa (2003), “Probabilistic Programming”, Chapter 5 in “Stochastic Programming”)

Applications in optimization

Theorem (Convexity of Upper Level Sets)

Let $f : S \to \mathbb{R}$ be a quasi-concave function, where $S \subseteq \mathbb{R}^n$ is a convex set. Then for any $t \in \mathbb{R}$, the upper level set

$$U_t = \{\boldsymbol{x} \in S \mid f(\boldsymbol{x}) \geq t\}$$

is convex.

Corollary (Feasible Set Convexity)

If f is quasi-concave, then the constraint:

$$f(\boldsymbol{x}) \geq t$$

*defines a **convex feasible set** for any $t \in \mathbb{R}$.*

Applications in stochastic programming

For chance-constrained programming problems of the form:

$$P[g(\boldsymbol{x}, \boldsymbol{\xi}) \leq 0] \geq 1 - \alpha,$$

if $P[g(\boldsymbol{x}, \boldsymbol{\xi}) \leq 0]$ is quasi-concave in $\boldsymbol{x}$, then $\{\boldsymbol{x} \mid P[g(\boldsymbol{x}, \boldsymbol{\xi}) \leq 0] \geq 1 - \alpha\}$ is **convex**: with a convex objective, the problem is a convex program.

Convexity is not tractability, though: evaluating the probability and its gradient still requires multivariate integration.

Log-concavity

Definition (Log-Concave Function)

A function $f : S \to \mathbb{R}_+$, where $S \subseteq \mathbb{R}^n$ is a convex set, is **log-concave** if for all $\boldsymbol{x}, \boldsymbol{y} \in S$ and $\lambda \in [0, 1]$:

$$f(\lambda \boldsymbol{x} + (1 - \lambda)\boldsymbol{y}) \geq f(\boldsymbol{x})^{\lambda} f(\boldsymbol{y})^{1-\lambda}$$

A log-concave function is quasi-concave, as $f(\boldsymbol{x})^{\lambda} f(\boldsymbol{y})^{1-\lambda} \geq \min\{f(\boldsymbol{x}), f(\boldsymbol{y})\}$.

A probability measure P is **log-concave** if $P[\lambda A + (1 - \lambda)B] \geq P[A]^{\lambda} P[B]^{1-\lambda}$ for all convex sets A, B and $\lambda \in [0, 1]$; this holds as soon as P has a log-concave density (Prékopa, 1973).

Jointly Convex Constraint Function

Definition

A constraint function $g : \mathbb{R}^n \times \mathbb{R}^p \to \mathbb{R}$ is **jointly convex** in $(\boldsymbol{x}, \xi)$ if for all $(\boldsymbol{x}_1, \xi_1), (\boldsymbol{x}_2, \xi_2) \in \mathbb{R}^n \times \mathbb{R}^p$ and all $\lambda \in [0, 1]$:

$$g(\lambda \boldsymbol{x}_1 + (1-\lambda)\boldsymbol{x}_2, \lambda \xi_1 + (1-\lambda)\xi_2) \leq \lambda g(\boldsymbol{x}_1, \xi_1) + (1-\lambda) g(\boldsymbol{x}_2, \xi_2)$$

Prékopa's fundamental theorem

Theorem

Let $\boldsymbol{\xi} \in \mathbb{R}^p$ be a random vector with log-concave probability distribution, and let $g : \mathbb{R}^n \times \mathbb{R}^p \to \mathbb{R}$ be jointly quasi-convex in $(\boldsymbol{x}, \boldsymbol{\xi})$. Then the probability function

$$\phi(\boldsymbol{x}) = P[g(\boldsymbol{x}, \boldsymbol{\xi}) \leq 0]$$

is log-concave (and hence quasi-concave) in $\boldsymbol{x}$.

(Prékopa, A. (1973). On logarithmic concave measures and functions. Acta Scientiarum Mathematicarum 34, 335–343)

Examples of jointly quasi-convex functions

Example (Affine in $(\boldsymbol{x}, \boldsymbol{\xi})$)

If $g(\boldsymbol{x}, \boldsymbol{\xi}) = \mathbf{a}^\top \boldsymbol{x} + \mathbf{b}^\top \boldsymbol{\xi} + c$, where $\mathbf{a} \in \mathbb{R}^n$, $\mathbf{b} \in \mathbb{R}^p$, $c \in \mathbb{R}$, then g is jointly quasi-convex (in fact, jointly convex).

Example (Maximum)

The maximum of quasi-convex functions is quasi-convex.

Common log-concave distributions

$$\text{Normal:} \quad f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

$$\text{Exponential:} \quad f(x) = \lambda \exp(-\lambda x), \quad x \geq 0$$

$$\text{Uniform on convex } S: \quad f(x) = \frac{1}{\mathrm{vol}(S)}, \quad x \in S$$

$$\text{Gamma (shape } k \geq 1\text{):} \quad f(x) = \frac{\lambda^k}{\Gamma(k)} x^{k-1} e^{-\lambda x}, \quad x > 0,\ \lambda > 0$$

$$\text{Beta } (a \geq 1, b \geq 1)\text{:} \quad f(x) = \frac{x^{a-1}(1-x)^{b-1}}{B(a,b)}, \quad x \in [0,1]$$

$$\text{Multivariate normal:} \quad f(x) = \frac{1}{\sqrt{(2\pi)^n \det(\Sigma)}} \exp\left(-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)\right)$$

Solution methods for log-concave distributions

With such a distribution, the feasible set is convex, and you can

- use Lagrangian techniques
- use a reduced-gradient technique (see Kall & Wallace, Section 4.1)

Joint constraints

Theorem (Prékopa's Theorem for Joint Constraints)

If

1. ξ *has a log-concave probability distribution, and*
2. *the set* $\{(\boldsymbol{x}, \boldsymbol{\xi}) \mid g_i(\boldsymbol{x}, \boldsymbol{\xi}) \leq 0,\ i = 1, \ldots, m\}$ *is convex, for instance because each* g_i *is jointly quasi-convex in* $(\boldsymbol{x}, \boldsymbol{\xi})$*,*

then the function

$$\phi(\boldsymbol{x}) = P\left[g_i(\boldsymbol{x}, \boldsymbol{\xi}) \leq 0,\ i = 1, \ldots, m\right]$$

is log-concave (hence quasi-concave) in $\boldsymbol{x}$*, and thus the feasible set* $\{\boldsymbol{x} \mid \phi(\boldsymbol{x}) \geq 1 - \alpha\}$ *is convex.*

Maximum reformulation

For joint constraints $P[g_j(\boldsymbol{x}, \boldsymbol{\xi}) \leq 0,\ j = 1, \ldots, m] \geq 1 - \alpha$, define:

$$G(\boldsymbol{x}, \boldsymbol{\xi}) = \max_{j=1,\ldots,m} g_j(\boldsymbol{x}, \boldsymbol{\xi})$$

Then the constraint becomes:

$$P[G(\boldsymbol{x}, \boldsymbol{\xi}) \leq 0] \geq 1 - \alpha$$

If every g_j is jointly quasi-convex, so is G: Prékopa's theorem for a single constraint then covers joint constraints.

Further reading

For a more complete analysis of convexity in chance-constrained programming, see Section 4.2 in Lectures on Stochastic Programming: Modeling and Theory (3rd edition, Shapiro, Dentcheva, Ruszczyński).

Linear inequalities

Assume that the chance constraint can be written in the form

$$P[T(\xi)x \geq h(\xi)] \geq 1 - \alpha.$$

Theorem

Suppose $T(\xi) = T$ is fixed, and $h(\xi)$ has a quasi-concave probability measure P. Then $K_1(\alpha)$ is convex for $0 \leq \alpha \leq 1$.

Single constraint: easy case

- The situation in the single constraint case is sometimes simple.
- Suppose again that $T_i(\boldsymbol{\xi}) = T_i$ is constant. Then

$$P[T_i x \geq h_i(\boldsymbol{\xi})] = F_{h_i}(T_i x) \geq 1 - \alpha$$

so the deterministic equivalent is

$$T_i x \geq F_{h_i}^{-1}(1 - \alpha)$$

...linear constraint! The resulting problem is still linear: compared with requiring $T_i x \geq h_i(\xi)$ for every realization, the worst case of $h_i(\boldsymbol{\xi})$ is replaced by its $(1 - \alpha)$-quantile.

- Recall that the inverse of the cdf is defined as

$$F_{h_i}^{-1}(p) = \inf\{t : F_{h_i}(t) \geq p\}, \quad p \in (0, 1).$$

Other "solvable" cases

Consider a single constraint $\boldsymbol{\xi}^T x \geq h$, with $h \in \mathbb{R}$ fixed and $\boldsymbol{\xi} = (\xi_1, \xi_2, \ldots, \xi_n)$ following a multivariate normal distribution with mean μ and positive definite covariance matrix Σ. Then, for $x \neq 0$,

$$\frac{\sum_{i=1}^n \xi_i x_i - \mu^T x}{\sqrt{x^T \Sigma x}} \sim N(0,1),$$

and

$$K_1(\alpha) = \{x \mid \mu^T x \geq h + \Phi^{-1}(1-\alpha)\sqrt{x^T \Sigma x}\},$$

where Φ is the standard normal cdf.

$K_1(\alpha)$ is a convex set for $\alpha \leq 0.5$. For $\alpha < 0.5$, it can be expressed as a second-order cone constraint:

$$\|\Sigma^{1/2} x\|_2 \leq \frac{1}{\Phi^{-1}(1-\alpha)}(\mu^T x - h)$$

Second-order cone programming

A second-order cone program (SOCP) is a convex optimization problem of the form

$$\begin{aligned} \min_x \ & f^T x \\ \text{s.t. } & \|A_i x + b_i\|_2 \leq c_i^T x + d_i, \ i = 1, \ldots, m \\ & Fx = g \end{aligned}$$

where $x \in \mathbb{R}^n$, $f, c_i \in \mathbb{R}^n$, $A_i \in \mathbb{R}^{n_i \times n}$, $b_i \in \mathbb{R}^{n_i}$, $d_i \in \mathbb{R}$, $F \in \mathbb{R}^{p \times n}$, and $g \in \mathbb{R}^p$.

SOCPs can be solved efficiently by interior point methods.

Example: chance-constrained portfolio optimization

(Taken from S. Boyd and J. Linderoth)

- Suppose we want to invest in n assets, providing random return rates $\beta_1, \beta_2, \ldots, \beta_n$.
- $\boldsymbol{\beta} \sim N(\boldsymbol{\mu}, \Sigma)$.
- W: total budget to invest.
- Suppose that we want to ensure a return of at least b. We cannot guarantee it all the time, but we want it to occur most of the time.

Example: chance-constrained portfolio optimization (cont'd)

Let $x_i \geq 0$ be the amount invested in asset i. Constraints:

$$P\left[\sum_{i=1}^{n} \beta_i x_i \geq b\right] \geq 1 - \alpha \tag{5}$$

$$\sum_{i=1}^{n} x_i \leq W$$

$$x_i \geq 0, \ i = 1, \ldots, n.$$

If $b < 0$, (5) bounds the loss $-\beta^T x$ by $-b$ with probability at least $1 - \alpha$: a Value-at-Risk constraint (Ruszczyński and Shapiro, 2003).

Example: deterministic equivalent of the chance constraint

The return $\beta^T x$ is normal, with mean $\mu^T x$ and variance $x^T \Sigma x$. For $x \neq 0$ and Σ positive definite,

$$Z = \frac{\beta^T x - \mu^T x}{\sqrt{x^T \Sigma x}} \sim N(0,1).$$

Standardizing, then using the symmetry of $N(0,1)$, $1 - \Phi(-t) = \Phi(t)$:

$$P\left[\beta^T x \geq b\right] = P\left[Z \geq \frac{b - \mu^T x}{\sqrt{x^T \Sigma x}}\right] = 1 - \Phi\left(\frac{b - \mu^T x}{\sqrt{x^T \Sigma x}}\right) = \Phi\left(\frac{\mu^T x - b}{\sqrt{x^T \Sigma x}}\right).$$

As Φ is continuous and increasing, (5) holds if and only if

$$\frac{\mu^T x - b}{\sqrt{x^T \Sigma x}} \geq \Phi^{-1}(1-\alpha) \quad \Longleftrightarrow \quad \mu^T x - \Phi^{-1}(1-\alpha)\sqrt{x^T \Sigma x} \geq b.$$

For $\alpha < 0.5$, $\Phi^{-1}(1-\alpha) > 0$, and this is a second-order cone constraint.

Example: chance-constrained portfolio optimization (cont'd)

We can also interpret x_i as the proportion of the portfolio invested in asset i, by normalizing $\sum_{i=1}^{n} x_i = 1$. Here b is the minimum return rate of the portfolio and x is the portfolio allocation vector.

We can add constraints on x to ensure diversification, summarized by requiring $x \in \mathcal{C}$.

A complete program can now be expressed as:

$$\begin{aligned} \max_x \ & \mathbb{E}[\beta^T x] = \mu^T x \\ \text{s.t. } & P\left[\beta^T x \geq b\right] \geq 1 - \alpha \\ & \sum_{i=1}^{n} x_i = 1 \\ & x \in \mathcal{C} \end{aligned}$$

Example: loss constraint

Setting b to 0 means that we want to ensure that we will not suffer a loss with high probability. Typically, α is set to 0.1, 0.05, 0.01,...

As $\beta^T x$ has a continuous distribution, the chance constraint can also be expressed as

$$P\left[\beta^T x \leq 0\right] \leq \alpha.$$

We can also allow short selling by permitting some x_i to be negative.

Numerical illustration

(Taken from S. Boyd – `http://ee364a.stanford.edu/lectures/chance_constr.pdf`)
$n = 10$ assets, $\alpha = 0.05$,$\mathcal{C} = \{x \mid x \succeq -0.1\}$

Compare:

- optimal portfolio
- optimal portfolio without loss risk constraint
- uniform portfolio $(1/n)\mathbf{1}$

portfolio	$\mathbb{E}[\beta^T x]$	$P[\beta^T x \leq 0]$
optimal	7.51	5.0%
w/o loss constraint	10.66	20.3%
uniform	3.41	18.9%

Short selling case

Let's ignore the non-negativity and budget constraints and consider the program (Ruszczyński and Shapiro, 2003)

$$\begin{aligned} \min_x \ & -\mu^T x \\ \text{s.t. } & z_\alpha \sqrt{x^T \Sigma x} - \mu^T x + b \leq 0, \end{aligned}$$

with $z_\alpha = \Phi^{-1}(1-\alpha) > 0$, i.e., $\alpha < 0.5$.
The Lagrangian is

$$\begin{aligned} L(x, \lambda) &= -\mu^T x + \lambda \left(z_\alpha \sqrt{x^T \Sigma x} - \mu^T x + b \right) \\ &= -(1+\lambda)\mu^T x + \lambda z_\alpha \sqrt{x^T \Sigma x} + \lambda b \end{aligned}$$

Reminder: KKT conditions

For $\min_x f(x)$ s.t. $g(x) \leq 0$, with f and g continuously differentiable and $L(x, \lambda) = f(x) + \lambda g(x)$, the Karush-Kuhn-Tucker (KKT) conditions are

$$\begin{aligned} \nabla_x L(x, \lambda) &= 0 && \text{(stationarity)} \\ g(x) &\leq 0 && \text{(primal feasibility)} \\ \lambda g(x) &= 0 && \text{(complementarity)} \\ \lambda &\geq 0 && \text{(dual feasibility)} \end{aligned}$$

- Necessary: a local minimizer satisfies them, for some λ, under a constraint qualification. For a convex problem, the Slater condition suffices: $g(\bar{x}) < 0$ for some $\bar{x}$.
- Sufficient for a convex problem: a KKT point is a global minimizer.

(See the KKT background notes.)

Short selling case: KKT conditions

$$\nabla_x L(x, \lambda) = 0, \qquad \lambda \geq 0,$$
$$z_\alpha \sqrt{x^T \Sigma x} - \mu^T x + b \leq 0, \qquad \lambda \left(z_\alpha \sqrt{x^T \Sigma x} - \mu^T x + b \right) = 0.$$

The problem is convex, as $z_\alpha > 0$ and $\sqrt{x^T \Sigma x} = \|\Sigma^{1/2} x\|_2$ is convex. If $b < 0$, $x = 0$ is strictly feasible (Slater condition): the KKT conditions are then necessary and sufficient.

The constraint is differentiable only where $x^T \Sigma x > 0$: $\sqrt{x^T \Sigma x}$ has a kink wherever it vanishes.

Short selling case: solving the KKT conditions

We have

$$\nabla_x L(x, \lambda) = -(1 + \lambda)\mu + \frac{\lambda z_\alpha \Sigma x}{\sqrt{x^T \Sigma x}}$$

If $\lambda = 0$,

$$\nabla_x L(x, \lambda) = 0 \implies \mu = 0.$$

Assuming $\mu \neq 0$, we must have $\lambda > 0$. Therefore:

$$z_\alpha \sqrt{x^T \Sigma x} - \mu^T x + b = 0$$

Short selling case: no risk-free asset

(Ruszczyński and Shapiro, 2003)

Without a risk-free asset, every portfolio $x \neq 0$ is risky: Σ is positive definite. Assume also $b < 0$: a loss of at most $-b$ is tolerated.

1. Direction. Let (x, λ) be an unknown KKT point. As $\lambda > 0$, stationarity gives

$$\Sigma x = t\,\mu, \quad \text{where the number } t = \frac{(1+\lambda)\sqrt{x^T \Sigma x}}{\lambda z_\alpha} > 0, \quad \text{so} \quad x = t\,\Sigma^{-1}\mu.$$

2. Return and risk along this direction. With $\rho = \sqrt{\mu^T \Sigma^{-1} \mu}$,

$$\mu^T x = t\rho^2, \qquad \sqrt{x^T \Sigma x} = t\rho.$$

3. Step length. The constraint is active, as $\lambda > 0$:

$$z_\alpha t\rho - t\rho^2 + b = 0 \quad \Longleftrightarrow \quad t\rho\,(z_\alpha - \rho) = -b.$$

Short selling case: no risk-free asset (cont'd)

As $-b > 0$, the step length equation has a solution $t > 0$ if and only if $\rho < z_\alpha$.

- $\rho < z_\alpha$: with $\sqrt{x^T \Sigma x} = t\rho$, the definition of t gives the multiplier: $t = (1+\lambda)t\rho/(\lambda z_\alpha) \iff \lambda = \rho/(z_\alpha - \rho) > 0$. All KKT conditions hold: the unique KKT point, hence the solution, is

$$x^* = \frac{-b}{\rho(z_\alpha - \rho)} \Sigma^{-1}\mu, \qquad \text{with expected return } \mu^T x^* = \frac{-b\,\rho}{z_\alpha - \rho}.$$

- $\rho \geq z_\alpha$: there is no KKT point. Along $x = t\,\Sigma^{-1}\mu$, the constraint value is $t\rho(z_\alpha - \rho) + b \leq b < 0$ for every $t \geq 0$, while $\mu^T x = t\rho^2 \to \infty$: the problem is unbounded.

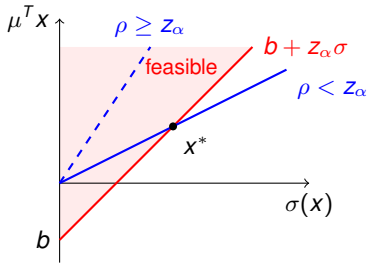
Short selling case: interpretation

Sharpe ratio (Sharpe, 1966): expected return per unit of risk, $S(x) = \mu^T x/\sigma(x)$, with $\sigma(x) = \sqrt{x^T \Sigma x}$ the standard deviation of the return (usually, of the return in excess of the risk-free rate). Scale-free: $S(tx) = S(x)$, $t > 0$.

Best Sharpe ratio. By Cauchy-Schwarz, $\mu^T x = (\Sigma^{-1/2}\mu)^T(\Sigma^{1/2}x) \leq \rho\,\sigma(x)$, with equality iff $x = t\,\Sigma^{-1}\mu$, $t \geq 0$: $\rho = \max_x S(x)$, attained in the KKT direction.

Chance constraint: $\mu^T x \geq b + z_\alpha \sigma(x)$, a safety margin of z_α standard deviations. Along $x = t\,\Sigma^{-1}\mu$, the return $\rho\,\sigma$ and the margin $z_\alpha\sigma$ grow linearly with σ:

- $\rho < z_\alpha$: the margin grows faster; scaling up stops at x^*.
- $\rho \geq z_\alpha$: the return grows at least as fast; nothing stops the scaling (the ignored budget constraint would).



Generalization

A more general form is

$$\begin{aligned} \min_x \; & h(x) \\ \text{s.t. } & P[g_1(x,\xi) \le 0, \ldots, g_r(x,\xi) \le 0] \ge 1-\alpha \\ & h_1(x) \le 0, \ldots, h_m(x) \le 0. \end{aligned}$$

Using $G(x,\xi) = \max_{j=1,\ldots,r} g_j(x,\xi)$, the chance constraint can be expressed via the indicator of feasibility:

$$\mathbb{E}\left[\mathbb{I}_{(-\infty,0]}(G(x,\xi))\right] \ge 1-\alpha, \quad \text{where} \quad \mathbb{I}_{(-\infty,0]}(t) = \begin{cases} 1 & \text{if } t \le 0, \\ 0 & \text{if } t > 0, \end{cases}$$

or equivalently via the violation indicator $\mathbb{I}_{(0,\infty)}$:

$$\mathbb{E}\left[\mathbb{I}_{(0,\infty)}(G(x,\xi))\right] \le \alpha, \quad \text{where} \quad \mathbb{I}_{(0,\infty)}(t) = \begin{cases} 1 & \text{if } t > 0, \\ 0 & \text{if } t \le 0. \end{cases}$$

Solution methods for the general case

- Usually very hard.
- Use a bounding approximation or sample average approximation (SAA).
- We will discuss it in more detail when introducing Monte Carlo techniques.

Probabilistic programming

Source: András Prékopa (2003), “Probabilistic Programming”, Chapter 5 in “Stochastic Programming”, A. Ruszczyński and A. Shapiro (editors), Elsevier.

- Sometimes we only want to maximize a probability.
- General form:

$$\max_x P[g_1(x,\xi) \leq 0, \ldots, g_r(x,\xi) \leq 0]$$
$$\text{s.t. } h_1(x) \leq 0, \ldots, h_m(x) \leq 0.$$

Measures of violation

- A chance constraint allows constraint violation with some probability.
- The violation can be large.
- It is often desirable to avoid too large violations.
- Can we penalize the violation?

Value at Risk

Source: `https://web.stanford.edu/class/ee364a/lectures/chance_constr.pdf`

Value-at-risk of a loss Z (large values are bad), at level $\eta \in (0, 1)$:

$$\text{VaR}(Z; \eta) = \inf\{\gamma \mid P[Z \leq \gamma] \geq \eta\}$$

Therefore, the value-at-risk is simply the inverse of the cdf evaluated at η:

$$\text{VaR}(Z; \eta) = F_Z^{-1}(\eta).$$

Conditional Value at Risk

$$\text{CVaR}(Z;\eta) = \inf_{\beta}\left(\beta + \frac{1}{1-\eta}\mathbb{E}\left[(Z-\beta)_+\right]\right).$$

Assume that the distribution of Z is continuous.

Solution β^* obtained by solving

$$0 = \frac{d}{d\beta}\left(\beta + \frac{1}{1-\eta}\mathbb{E}\left[(Z-\beta)_+\right]\right) = 1 - \frac{1}{1-\eta}P[Z \geq \beta],$$

leading to

$$P[Z \geq \beta^*] = 1-\eta \quad \Longleftrightarrow \quad P[Z \leq \beta^*] = \eta,$$

satisfied by $\beta^* = \text{VaR}(Z;\eta)$, and only by it if F_Z is strictly increasing.

Expected shortfall

Conditional tail expectation (or expected shortfall):

$$\begin{aligned}\mathbb{E}[Z \mid Z \geq \beta^*] &= \mathbb{E}[\beta^* + (Z - \beta^*) \mid Z \geq \beta^*] \\ &= \beta^* + \frac{\mathbb{E}\left[(Z - \beta^*)_+\right]}{P[Z \geq \beta^*]} \\ &= \mathrm{CVaR}(Z; \eta)\end{aligned}$$

- Can be added to the objective.
- Can be used as a constraint: *conditional expectation constraint*

$$\mathbb{E}[Z \mid Z \geq \beta^*] \leq d.$$

CVaR and chance constraints

In the continuous case, $\mathrm{CVaR}(Z;\eta)$ is the mean of Z beyond $\mathrm{VaR}(Z;\eta)$, hence $\mathrm{VaR}(Z;\eta) \leq \mathrm{CVaR}(Z;\eta)$ (true in general). With $Z = G(x,\xi)$ and $\eta = 1-\alpha$:

$$\mathrm{CVaR}(G(x,\xi);1-\alpha) \leq 0 \implies \mathrm{VaR}(G(x,\xi);1-\alpha) \leq 0 \iff P[G(x,\xi) \leq 0] \geq 1-\alpha.$$

By definition of the CVaR, the left-hand constraint holds if and only if, for some $\beta \in \mathbb{R}$,

$$\beta + \frac{1}{\alpha}\mathbb{E}\left[(G(x,\xi)-\beta)_+\right] \leq 0,$$

a convex constraint in (x,β) when $G(\cdot,\xi)$ is convex: a conservative convex approximation of the chance constraint. With finitely many scenarios and affine g_j, it is a set of linear constraints, with one auxiliary variable per scenario.

(Rockafellar and Uryasev, 2000; Nemirovski and Shapiro, 2006)

Integrated chance constraints

- Consider the stochastic constraints

$$g_i(x, \xi) \leq 0, \quad i = 1, \ldots, r.$$

- Integrated chance constraint:

$$\mathbb{E}\left[\max_{i=1,\ldots,r} (g_i(x, \xi))_+\right] \leq d.$$

- Unlike a chance constraint, it defines a convex set for any distribution of ξ, as soon as the g_i are convex in x. With a finite discrete distribution and affine g_i, this set is a polyhedron: the constraint can be handled by linear programming.

For more details, see Chapter 6 (Section 6.3 for joint constraints), Willem K. Klein Haneveld, Maarten H. van der Vlerk, Ward Romeijnders (2020), “Stochastic Programming - Modeling Decision Problems Under Uncertainty”, Springer.